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Appendices

A OPEN SOURCE MODELS

Apart from PaLM 2 [46] models, we have also experimented with open source models such as Llama 2 [48] and Llama 3. These finetuned models are publicly available on Hugging Face. The finetuning scripts are available at: <https://github.com/google/speaker-id/tree/master/DiarizationLM/unsloth>.

A.1 google/DiarizationLM-13b-Fisher-v1

Model URL: <https://huggingface.co/google/DiarizationLM-13b-Fisher-v1>
Foundation model: <https://huggingface.co/unsloth/llama-2-13b-bnb-4bit>

This model is finetuned on the training subset of the Fisher corpus, using a LoRA adapter [63] of rank 256. The total number of training parameters is 1,001,390,080. With a batch size of 16, this model has been trained for 12000 steps, which is ~ 4 epochs of the training data.

We use the “mixed” flavor during our training, meaning we combine data from “hyp2ora” and “deg2ref” flavors (see Section 3.5 for the definition of these flavors). After the prompt builder, we have a total of 48,142 prompt-completion pairs in our training set.

The finetuning took more than 3 days on a Google Cloud VM instance that has one NVIDIA A100 GPU with 80GB memory.

The maximal length of the prompt to this model is 6000 characters, including the “`_-->_`” suffix. The maximal sequence length is 4096 tokens.

The evaluation results of this model is also provided in Table 1. We can see that similar to the finetuned PaLM 2-S models, this model also significantly reduces the WDER and cpWER on the Fisher testing set. However, on the Callhome testing set, we are seeing degradation in WDER and cpWER, indicating overfitting on the Fisher dataset. We have found this is because the loss is computed on both the prompt and the completion tokens during training.

A.2 google/DiarizationLM-8b-Fisher-v1

Model URL: <https://huggingface.co/google/DiarizationLM-8b-Fisher-v1>
Foundation model: <https://huggingface.co/unsloth/llama-3-8b-bnb-4bit>

This model is finetuned on the training subset of the Fisher corpus, using a LoRA adapter [63] of rank 256. The total number of training parameters is 671,088,640. With a batch size of 16, this model has been trained for 25400 steps, which is ~ 8 epochs of the training data.

Other configurations of this model is the same as the model in the previous section. Note that we still use a maximal sequence length of 4096 tokens even if Llama 3 supports longer sequence.

The finetuning took more than 4 days on a Google Cloud VM instance that has one NVIDIA A100 GPU with 80GB memory.

The evaluation results of this model is also provided in Table 1. We can see that its performance is significantly worse than the Llama 2 13B model from Appendix A.1, but still better than the USM + turn-to-diarize baseline on Fisher testing set. Similarly, the performance on Callhome is worse than baseline, because the loss is computed on both the prompt and the completion tokens during training.

A.3 google/DiarizationLM-8b-Fisher-v2

Model URL: <https://huggingface.co/google/DiarizationLM-8b-Fisher-v1>
Foundation model: <https://huggingface.co/unsloth/llama-3-8b-bnb-4bit>

This model is finetuned on the training subset of the Fisher corpus, using a LoRA adapter [63] of rank 256. The total number of training parameters is 671,088,640. With a batch size of 16, this model has been trained for 28800 steps, which is ~ 9 epochs of the training data.

Note that when training this model, **the loss is only computed on the completion tokens**. This is the biggest different between this model and the previous model in Appendix A.2 (google/DiarizationLM-8b-Fisher-v1).

Other configurations of this model is the same as the model in the previous section. Note that we still use a maximal sequence length of 4096 tokens even if Llama 3 supports longer sequence.

The finetuning took more than 4 days on a Google Cloud VM instance that has one NVIDIA A100 GPU with 80GB memory.

The evaluation results of this model is also provided in Table 1. We can see that its performance is significantly better than the model from Appendix A.2. Specifically, we no longer see degradation on Callhome testin set.

B OTHER DIARIZATION CAPABILITIES OF LLMs

In our studies, we have found multiple diarization capabilities of LLMs. Due to the lack of relevant training and testing datasets, this paper mainly focuses on evaluating how LLMs can reduce diarization errors. In this section, we provide several examples to demonstrate LLM’s other diarization capabilities. All the provided examples are obtained by directly sending the prompts to Google Bard [36]⁴.

B.1 AUTOFILLING SPEAKER NAMES

In real world applications, real speaker names or aliases are much more readable than anonymous speaker labels such as “speaker 1” or “speaker 2”. In many cases, speaker names can be obtained from the context of the conversation and from additional metadata of the conversation. In meeting summarization, the names can be obtained from the list of meeting attendants. And for video captioning, the names may be extracted from the video title or description. An example is provided below.

Prompt:

[video title] Tom meets Patrick in the parking lot
[video description] A short conversation

Replace the <spk:?> tokens from the below video caption with actual person names.

<spk:1> Good morning Patrick, how are you?
<spk:2> Good, good. How are you Tom?
<spk:1> Pretty good. Going to work?
<spk:2> Yes. Busy day. How are your kids? Do they go to school?
<spk:1> Oh they are too young for that. I sent them to daycare earlier today.
<spk:2> Oh yeah I forgot about that.

Completion:

⁴We used an internal version of Bard that is based on a larger model and supports more tokens than the public version.